

Tailwind CSS in Frappe

Step-by-step setup guide for Frappe custom apps

Tailwind v4 · Frappe Framework · Production-ready

Overview

This guide documents the working setup for integrating Tailwind CSS v4 into a Frappe custom app. It covers installation, build configuration, and loading the stylesheet on a specific page using `frappe.require()`.

Approach	No Internet	Race Condition	Production Safe	Best For
CDN Script	✗ Broken	⚠ Possible	✗ No	Quick prototypes
Local Bundle (hooks.py)	✓ Works	✓ None	✓ Yes	Global styles (all pages)
Local Bundle (frappe.require)	✓ Works	✓ None	✓ Yes	Page-specific styles ✓

✓ Recommended approach: Local Bundle with `frappe.require()` — works offline, no race conditions, loads only on pages that need it.

Prerequisites

Before starting, ensure the following are available on your system:

Requirement	Details
Node.js + npm	Required to install and run Tailwind CLI
Frappe Bench	Your app must be inside a frappe-bench setup
Custom Frappe App	e.g. <code>custom_app</code> — the app you are adding Tailwind to
bench CLI	Used to build and sync assets

Installation Steps

Step 1 — Initialise npm in your app

Navigate to your app root and initialise a `package.json` file:

```
cd ~/frappe/frappe-bench/apps/custom_app
npm init -y
```

i The app root is the outer folder (e.g. apps/custom_app), NOT the inner Python package (apps/custom_app/custom_app).

Step 2 — Install Tailwind CSS v4

Install Tailwind and the CLI package:

```
npm install tailwindcss @tailwindcss/cli
```

⚠ This installs Tailwind v4. The setup is different from v3 — there is no tailwind.config.js and no npx tailwindcss init command.

Step 3 — Create the source CSS file

Create the input CSS file that Tailwind will compile from. The path below is relative to your app root:

```
# File: apps/custom_app/custom_app/public/css/tailwind.src.css
```

Contents of tailwind.src.css (v4 syntax):

```
@import "tailwindcss";

@source "../../custom_app/public/js/**/*.js";
@source "../../custom_app/public/**/*.html";
@source "../../custom_app/templates/**/*.html";
@source "../../custom_app/www/**/*.html";
```

i The @source directives tell Tailwind v4 which files to scan for class names. Without these, the output CSS will be almost empty.

⚠ Do NOT use @tailwind base / @tailwind components / @tailwind utilities — those are v3 syntax and will cause errors in v4.

Step 4 — Add build scripts to package.json

Open apps/custom_app/package.json and add the following scripts:

```
{
  "scripts": {
    "build:css": "tailwindcss -i ./custom_app/public/css/tailwind.src.css -o ./custom_app/public/css/tailwind.css --minify",
    "watch:css": "tailwindcss -i ./custom_app/public/css/tailwind.src.css -o ./custom_app/public/css/tailwind.css --watch"
  }
}
```

i Always write the command as a single line in package.json. Multi-line shell commands with backslash (\) do not work reliably here.

Step 5 — Build the CSS

Run the build command from inside the app root:

```
cd ~/frappe/frappe-bench/apps/custom_app
npm run build:css
```

This generates: `apps/custom_app/custom_app/public/css/tailwind.css`

To verify the correct classes were generated:

```
grep "p-6|max-w|rounded|text-xl" ~/frappe/frappe-bench/sites/assets/custom_app/css/tailwind.css
```

☒ During development, use `npm run watch:css` to auto-rebuild whenever you add new Tailwind classes to your JS files.

Step 6 — Sync assets with Frappe

After building, run bench build to copy the CSS into the sites/assets folder where Frappe can serve it:

```
cd ~/frappe/frappe-bench
bench build --app custom_app
```

The compiled file will be available at:

```
~/frappe/frappe-bench/sites/assets/custom_app/css/tailwind.css
```

Step 7 — Load CSS in your Frappe page

Use `frappe.require()` with a callback in your page JS file. This ensures the HTML renders only after the CSS is fully loaded, preventing flash of unstyled content:

```
// apps/custom_app/custom_app/public/js/sample-frappe-page.js

frappe.pages['sample-frappe-page'].on_page_load = function (wrapper) {
  var page = frappe.ui.make_app_page({
    parent: wrapper,
    title: 'Sample Frappe Page',
    single_column: true
  });

  frappe.require("/assets/custom_app/css/tailwind.css", function() {
    $(page.main).empty().html(`
      <div class="p-6 max-w-4xl mx-auto">
        <div class="bg-white rounded-2xl shadow p-6">
          <h2 class="text-xl font-semibold text-gray-800 mb-4">
            Welcome
          </h2>
          <p class="text-gray-500">Sample Frappe Page</p>
        </div>
      </div>
    `);
  });
}
```

```

    });
  });
}

```

☑ `frappe.require()` with a callback solves the race condition. The HTML is only injected after the CSS has finished loading.

Tailwind v3 vs v4 — Key Differences

You installed Tailwind v4 which has significant breaking changes from v3. Use this table as a reference:

Feature	Tailwind v3	Tailwind v4
Config file	<code>tailwind.config.js</code> ☑	Not needed ✗
CSS directive	<code>@tailwind base/components/utilities</code>	<code>@import "tailwindcss"</code>
Content scan	Manual in config	Auto-detected + <code>@source</code>
Init command	<code>npx tailwindcss init</code>	Does not exist ✗

Final File Structure

After completing all steps, your app should look like this:

```

apps/custom_app/
├── package.json           ← npm scripts (build:css, watch:css)
├── node_modules/         ← Tailwind + CLI (auto-generated)
├── package-lock.json
├── custom_app/           ← Python package (inner folder)
│   ├── hooks.py
│   ├── __init__.py
│   └── public/
│       ├── css/
│       │   ├── tailwind.src.css ← You write this (source)
│       │   └── tailwind.css     ← Auto-generated (served)
│       └── js/
│           └── sample-frappe-page.js ← Your page JS

```

Quick Reference — Daily Commands

Task	Command
Build CSS once	<code>npm run build:css</code>
Watch for changes (dev)	<code>npm run watch:css</code>
Sync assets to Frappe	<code>bench build --app custom_app</code>
Verify classes in output	<code>grep "p-6" sites/assets/custom_app/css/tailwind.css</code>
Hard refresh browser	<code>Ctrl + Shift + R</code>

Troubleshooting

Problem	Fix
Styles not applying on first load	Use <code>frappe.require()</code> with callback — never inject HTML before CSS loads
CSS output is almost empty	Add <code>@source</code> directives in <code>tailwind.src.css</code> pointing to your JS/HTML files
<code>npx tailwindcss init</code> fails	v4 has no init command — skip it, config file is not needed
<code>@tailwind base</code> throws error	Replace with <code>@import "tailwindcss"</code> — v3 directives don't work in v4
Styles still wrong after build	Run <code>bench build --app custom_app</code> then hard-refresh browser (Ctrl+Shift+R)
Frappe styles overriding Tailwind	Use more specific CSS selectors or add Tailwind's <code>!important</code> variant